



TECHNICAL UNIVERSITY OF BERLIN
RESEARCH LAB: MODELING OF COGNITIVE PROCESSES

Project Report

Agent-Based Modeling of Cyclical Fashion Trends with Population Dynamics

Written by
Thomas Hetzner

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1 Introduction

Clothing is a concept with roots reaching back many millennia. Researchers suggest that clothes may have been worn up to 75,000 years ago, based on the genetic analysis of the body louse [1]. Other early evidence includes preserved clothing remains found in the Italian cave of Arene Candide [2]. Although clothing has been a basic human need for such a long time, it is difficult to determine exactly when fashion first emerged and what its origins are [3]. Fashion is closely tied to trends, which can be understood from the perspective of Everett Rogers’ theory of the diffusion of innovations. Rogers classifies adopters into five distinct groups based on their receptiveness to change: innovators, early adopters, early majority, late majority, and laggards. Within this framework, momentum for a trend builds as it spreads from a small group of risk-takers to the broader social system. [4]. Applied specifically to the apparel industry, this process is known as the fashion cycle. A style is first adopted by trendsetters and by the time laggards join in, trendsetters have usually already moved on to the next new thing. This makes the older style fall out of fashion, and the cycle begins all over again with a new trend. While a specific trend might eventually lose all its followers, it often lies dormant, ready to be revived in the future [5].

The lifespan of a trend varies greatly. In the 20th century, trends often spanned over longer periods, sometimes defining entire decades, however, especially in fast fashion, the span short periods [6]. Since then, many of these past styles have experienced a revival, driven by nostalgia and the rising popularity of second-hand shopping [7]. However, the nature of fashion has shifted dramatically. In recent years, fueled by fast fashion, technological advances, and the heavy influence of social media, micro-trends arise at a rapid pace and are in fashion and then out fashion within the blink of an eye causing highly frequent trend switching behaviour. Because of this, multiple trends now coexist at the same time, turning modern fashion into a highly fragmented and fast-moving landscape [8].

Attempts to explain the complex dynamics behind these trends draw from a variety of different disciplines. Therefore, it is interesting whether a simple computational model is capable of capturing these shifting dynamics. In this project, an agent-based model is initially developed to simulate the cyclical nature of fashion trends. Since this initial model operates in a fixed environment, it remains unclear whether its mechanisms hold true in a scenario with population dynamics. Therefore, a model is built to reflect demographic transitions and merged with the fashion trend model to achieve a more realistic representation. The code for this model is available via GitHub.¹

¹<https://github.com/hetznero/ftdtm>

2 Methods

In the underlying project three agent-based models are developed. This section details the mathematical formulations and design choices for a foundational fashion trend model, a demographic transition model, and a unified framework integrating both systems.

2.1 Agent-based Fashion Trend Model

Fashion trends exhibit complex lifecycles characterized by initial emergence, widespread adoption, gradual decline, and occasional revival [5]. To simulate these dynamics, an agent-based model is formulated, in which human agents interact with a dynamic set of competing trends. The core behavioral mechanic is grounded in Everett Rogers’ theory of the diffusion of innovations [4], differentiating agents by their propensity to adopt trends based on current popularity.

The model consists of two primary interacting classes, the human agents and fashion trends. Each human agent i possesses an innate target popularity parameter drawn from a standard normal distribution, $P_{target,i} \sim \mathcal{N}(0, 1)$. This parameter dictates an agent’s position on the adoption curve, as trendsetters seek exclusivity (lower target popularity), whereas mass-market adopters prefer conformity (higher target popularity).

Each trend j is assigned an time-invariant quality parameter $q_j \sim \mathcal{U}(0.6, 1.4)$. This parameter serves as an abstraction for multiple real world properties, such as functional utility, durability, or accessibility of the trend. Additionally, a trend possesses an age variable $a_j(t)$, a latent energy level $E_j(t)$, and an active appeal $A_j(t)$. The global popularity of a trend, $P_j(t)$, is dynamically defined as the fraction of the total human population currently adopting it. These parameters are used to determine whether an agent wants to stick with its current trend or switch to another trend to maximize a utility function U , which is explained in the following.

Agent Decision Mechanics

At each time step, human agents reevaluate their choices with a probability of $p_{update} = 0.05$. When updating, an agent evaluates all currently active trends to maximize a utility function:

$$U_{i,j}(t) = A_j(t) - \omega |P_j(t) - P_{target,i}| + \xi_i(t)$$

where $\omega = 2.0$ acts as a penalty weight for deviating from the agent’s target popularity, and $\xi_i(t) \sim \mathcal{N}(0, 0.1)$ introduces noise to account for irrational or random choices. Agents switch to the trend j that maximizes $U_{i,j}(t)$.

Trend Lifecycle and Revival Mechanics

Trends exist in one of three states namely Active, Passive, or Dead. For Active trends, the age $a_j(t)$ increments at each time step. To model the natural lifecycle from initial hype to eventual fatigue, the active appeal $A_j(t)$ is calculated using a Gaussian function scaled by its quality:

$$A_j(t) = q_j \exp \left(-\frac{1}{2} \left(\frac{a_j(t) - \mu_q}{\sigma_q} \right)^2 \right)$$

where $\mu_q = 80q_j$ defines the peak time and $\sigma_q = 30q_j$ defines the temporal spread. This formulation ensures that higher-quality trends achieve a higher maximum appeal and sustain their cultural relevance for longer durations.

When an Active trend surpasses its peak age (μ_q) and its appeal drops below a critical threshold ($A_j(t) < 0.1$), it transitions into a Passive state. This represents a period where the trend is no longer actively adopted but remains in the cultural memory. During this state, a drift-diffusion model governs the trend's latent energy:

$$E_j(t+1) = E_j(t) + \alpha(q_j - 1) + \eta_j(t)$$

where $\alpha = 0.05$ is the base drift rate and $\eta_j(t) \sim \mathcal{N}(0, 0.05)$ is Gaussian noise. This stochastic process relies on the quality parameter q_j : high-quality trends ($q_j > 1$) exhibit a positive expected drift, making them more likely to be revived. If $E_j \geq 1.0$, the trend re-enters the Active state, resetting its age and energy to simulate a revival. Conversely, if $E_j \leq -1.0$, the trend is classified as Dead and permanently removed from the simulation.

Simulation Dynamics and Initialization

The global simulation operates over $T = 1200$ discrete time steps. It is initialized with a fixed population of $N = 1000$ agents and an initial set of $M_0 = 5$ active trends. To simulate the accelerating pace of the fashion industry driven by technology and globalization, new trends spawn dynamically throughout the simulation. The emergence probability $p_{spawn}(t)$ of a new trend at time t is governed by a base rate $p_{base} = 0.01$ and amplified by a logistic multiplier:

$$p_{spawn}(t) = p_{base} \left(1 + \frac{10}{1 + \exp(-0.02(t - t_{mid}))} \right)$$

where $t_{mid} = 600$ represents the midpoint of the simulation. This curve is similar to the one illustrated in Figure 1, but scaled differently.

As formulated above, this foundational fashion model operates within a fixed environment, assuming a static population of human agents. To incorporate the generational shifts and changing demographic structures influence into the behaviour of trend

dynamics, these dynamics are integrated via a dynamic population model, which is described in the following subsection.

2.2 Agent-based Demographic Transition Model

The demographic transition model describes the shift in population dynamics from high birth and death rates to low birth and death rates as a society develops [9]. To replicate these population shifts and the resulting transformations in age structure, another agent-based model is introduced.

A population of $N(t)$ autonomous human agents is simulated over a period of $T = 300$ discrete, yearly time steps. Each agent i is characterized by its age $a_i(t)$ and biological sex $s_i \in \{0, 1\}$, where $s_i = 0$ denotes female and $s_i = 1$ denotes male. At each time step, agents age by one year and are subject to stochastic mortality and fertility rates. To capture the dynamics of real-world societal development, a global resource and technology level, $R(t) \in [0, 1)$, is defined. This variable acts as a proxy for technological progress, medical advancements, and increased education, governing both mortality and fertility rates. This progress is modeled using a sigmoid function, which is illustrated in Figure 1, centered at the simulation midpoint:

$$R(t) = \frac{1}{1 + e^{-k(t-t_0)}}$$

where $k = 0.05$ determines the rate of technological adoption and $t_0 = 150$ represents the transition midpoint.

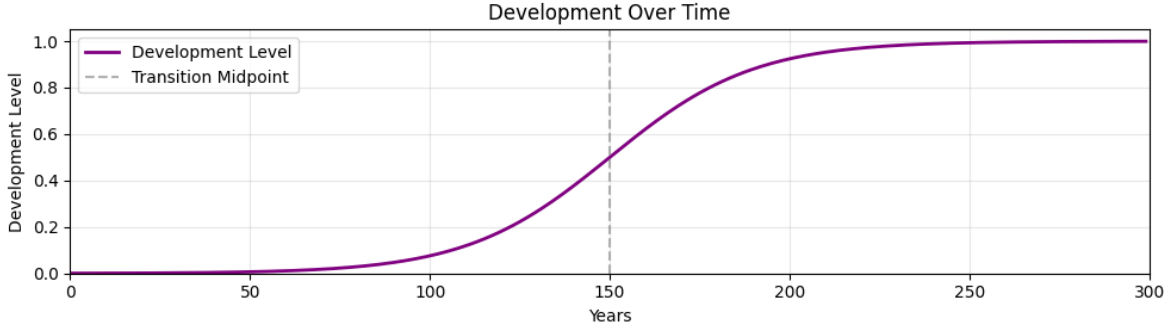


Figure 1: Development level over time modeled by a sigmoid function.

Mortality Mechanics

The probability of death for an agent increases exponentially with age, following a simplified Gompertz–Makeham law of mortality [10]. The base mortality probability is defined as $p_{base,d}(a_i) = 0.015e^{0.04a_i}$. Technological progress influences this risk, reflecting advancements in healthcare and living standards. The realized probability of death for

an agent at time t is modeled as:

$$p_d(a_i, t) = p_{base,d}(a_i)[1 - c_d R(t)]$$

where $c_d = 0.85$ is a scaling factor representing the maximum possible mortality reduction. Agents exceeding a hard maximum age threshold of 100 years are removed from the simulation.

Fertility Mechanics

Reproduction is restricted to female agents ($s_i = 0$) within a defined reproductive window of $15 \leq a_i \leq 45$ years. The base probability of an eligible agent giving birth in a given year is $p_{base,b} = 0.2$. In accordance with DTM theory, increased societal development (which correlates with higher education and family planning) leads to a decline in birth rates. Thus, the realized probability of giving birth is reduced by the resource level:

$$p_b(t) = p_{base,b}[1 - c_b R(t)]$$

where $c_b = 0.65$ represents the maximum reduction in fertility driven by societal progress.

Simulation Dynamics and Initialization

The simulation is initialized with $N(0) = 1000$ agents. To accurately represent a pre-transition society, initial agent ages are drawn from an exponential distribution, $a_i(0) \sim \text{Exp}(\lambda = 1/20)$, bounded at a maximum of 80 years. This initialization naturally yields a heavily young demographic structure. Over the 300-year simulation, the interaction between the Gompertz-Makeham mortality laws and the sigmoidal resource growth successfully transitions the population into an aging distribution, replicating the typical demographic dynamics of the modern Western world.

2.3 Agent-based Trend Model with Population Dynamics

To investigate the interplay between demographic shifts and cultural cycles, the previously described models are integrated into a unified framework. This fused model couples the population dynamics of the DTM with the mechanics of the fashion trend model.

Timescale Integration

A critical design choice in fusing these systems is the resolution of differing temporal scales. Fashion trends evolve rapidly, whereas demographic changes occur over generations. Therefore, the simulation time is discretized into quarters (four steps per year). The total simulation spans 300 years, yielding $T = 1200$ quarterly time steps. The fashion dynamics are updated at every quarterly step, While the demographic dynamics are updated annually at every fourth quarter.

Agent Fusion and Lifecycle Events

Human agents i are again instantiated as composite entities possessing both demographic traits (age $a_i(t)$ and sex s_i) and cultural preferences (target popularity $P_{target,i} \sim \mathcal{N}(0, 1)$). The dynamic nature of the population introduces new boundary conditions for the trend dynamics. When a mortality event occurs, the deceased agent is permanently removed from the simulation, and their currently adopted trend instantly loses a follower. Conversely, when a fertility event occurs, a new agent is introduced to the population at age zero. This new agent is immediately assigned a randomly drawn target popularity, continuously injecting novel consumer preferences into the market over time.

Dynamic Popularity and Trend Spawning

Because the total population size $N(t)$ fluctuates due to the DTM mechanics, the global popularity of a trend j can no longer be evaluated against a static denominator. Instead, it is dynamically computed at each quarter against the current living population:

$$P_j(t) = \frac{N_j(t)}{N(t)}$$

where $N_j(t)$ is the absolute number of living agents currently adopting trend j at quarter t .

Furthermore, the technological progress multiplier, which scales the probability of new trends spawning, is adapted to align with the quarterly simulation length. The emergence probability is defined as:

$$p_{spawn}(t) = p_{base} \left(1 + \frac{10}{1 + \exp(-0.02(t - t_{mid}))} \right)$$

where $p_{base} = 0.01$ and the inflection point is set to the simulation midpoint at $t_{mid} = 600$ quarters.

Simulation Dynamics and Initialization

The fused simulation is initialized with $M(0) = 5$ Trends and $N(0) = 1000$ agents, distributed exponentially by age to represent a pre-transition society, identical to the standalone DTM. This specific initialization choice allows the model to simulate the fashion trend dynamics with population dynamics.

3 Results

This section presents the results of the models described in Section 2, illustrated through a series of plots generated from the simulated models.

3.1 Agent-based Fashion Trend Model

The Agent-based Fashion Trend Model was simulated over 1,200 time steps, with each step representing one quarter, yielding a total simulation period of 300 years. Figure 2 illustrates the popularity of various fashion trends over time. It can be observed that, initially, fewer trends are present, with a single trend typically dominating the market for an extended period. In the later stages of the simulation, multiple trends coexist simultaneously, and their individual lifespans are notably shorter, driving a high-frequency trend-switching dynamic.

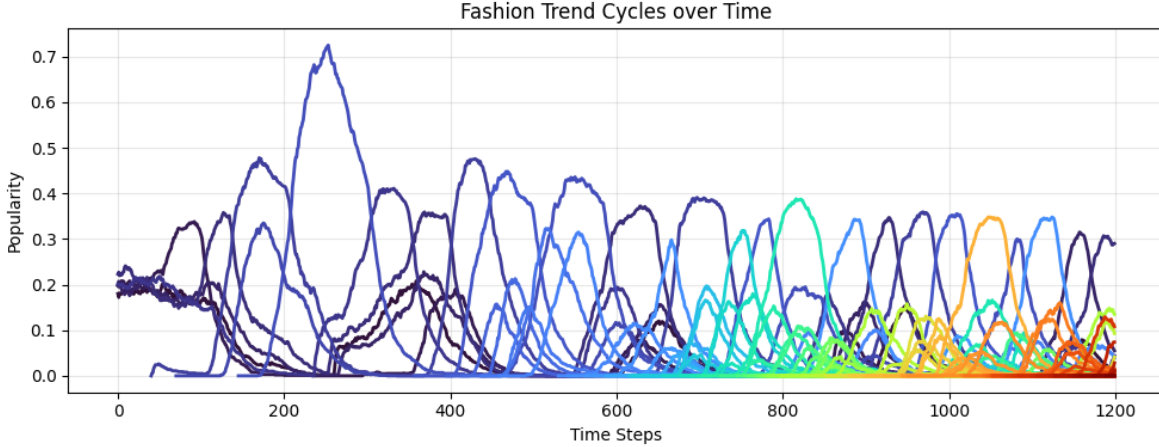


Figure 2: Fashion Trend Cycles over time of the Fashion Trend Model.

Figure 3 demonstrates that the drift-diffusion model, successfully enforces transitions either back into the active state or permanently into the dead state. While this plot captures categorical states, Figure 4 details the underlying drift of trends during their passive phase. This drift highlights how certain trends accumulate enough momentum to experience a revival, whereas others gradually fade and die.

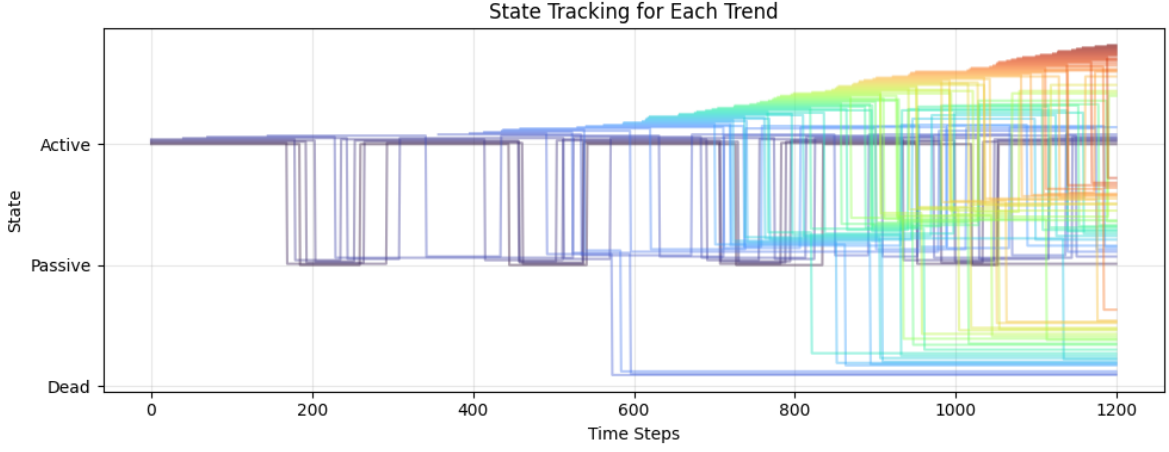


Figure 3: State Tracking of each Trend over time of the Fashion Trend Model (for better visibility of each trend, each state curve is slightly shifted).

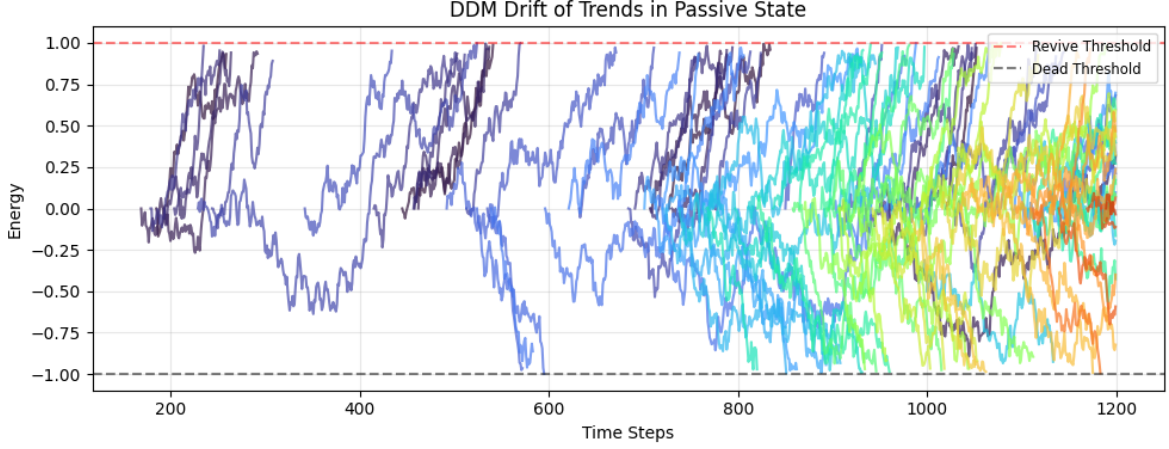


Figure 4: DDM Drift of Trends in Passive State over time of the Fashion Trend Model.

Finally, Figure 5 isolates Trends 0, 10, and 20 to better observe their dynamics. Trend 0 clearly completes more than one lifecycle, indicating a successful revival. Conversely, while Trends 10 and 20 achieve temporary popularity, combining their trajectories with their drift in Figure 4 suggests that both ultimately transition into the dead state.

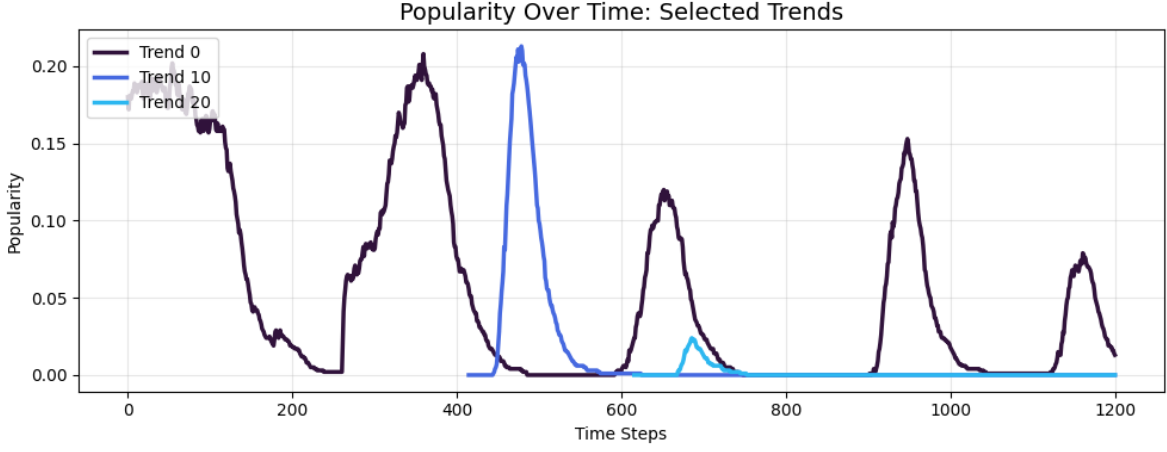


Figure 5: Popularity of selected Trends over time of the Fashion Trend Model.

3.2 Agent-based Demographic Transition Model

In Figure 6, the dynamics of the Agent-based Demographic Transition Model are illustrated. The simulation initializes with a population whose age distribution decays exponentially from 0 to 100 years. In these early stages, both birth and death rates are relatively high. Over time, as the societal development level, illustrated in 6, increases, these rates steadily decrease. The highly youthful structure of the initial population drives robust demographic momentum, leading to a continuous increase in the total population size that peaks at approximately 250 time steps, followed by a gradual decline. Notably, the death rate eventually surpasses the birth rate. This suggests a direct consequence of an increasingly aging population and aligns with the later stages of classical demographic transition theory. Furthermore, the birth and death rates fluctuate heavily at the beginning of the simulation, as the high volatility is likely due to stochastic noise inherent in the small initial population size ($N = 1000$) and gradually decreases as the total population increases.

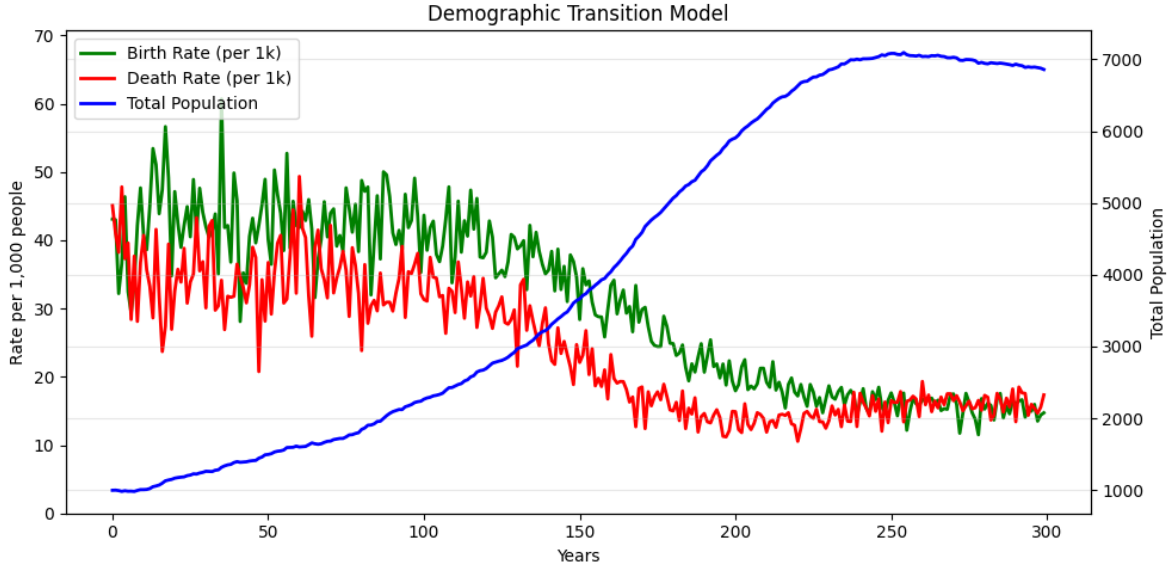


Figure 6: Demographic Rates and Total Population of the Demographic Transition Model.

Figure 7 presents the population's age distribution at time steps 0, 150, and 299. While these snapshots suggest that the model successfully undergoes a demographic transition, the continuous structural evolution is best observed in the animated distribution plotted in Figure 8. Over the course of the simulation, a distinct macro-level shift occurs, as the population smoothly transitions from an exponentially decaying distribution to a classic pyramidal shape. As the transition progresses, this pyramid rounds into a more bell-shaped curve, ultimately starting to shape more towards a aging population structure.

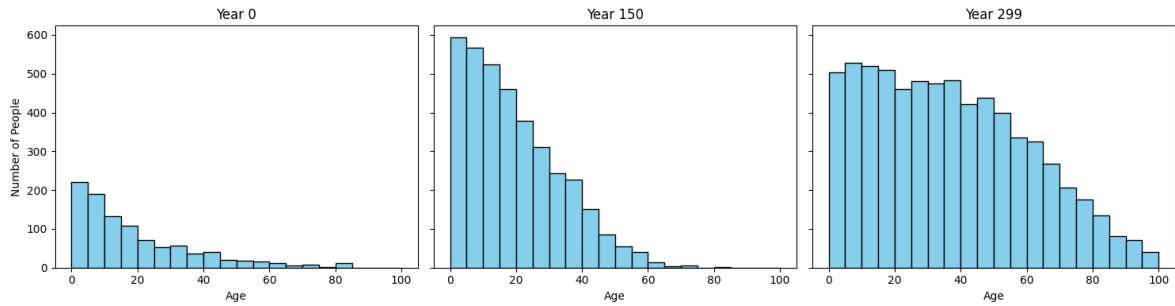


Figure 7: Population Overview of Year 0, Year 150 and Year 299 of the Demographic Transition Model.

Figure 8: Animated Plot of Population Dynamics over time

3.3 Agent-based Trend Model with Population Dynamics

Finally, the unified framework integrating the Agent-based Fashion Trend Model and the Demographic Transition Model was simulated. Because the core demographic mechanics remain unchanged, the fused model successfully maintains the population properties established previously, as illustrated in Figure 9. The primary point of interest is how these dynamic population shifts alter the cyclical behavior of fashion trends, which is captured in Figure 10. It is observable that, particularly in the beginning, the trend adoption curves are notably less smooth than in the static model. This increased volatility results directly from the continuous birth and death of agents, which rapidly injects and removes agents out of trends. However, in the later stages of the simulation, the trend behavior appears noticeably cleaner and more stable. This stabilization is likely due to the significantly larger overall population size achieved following the demographic boom, which dampens the stochastic noise of individual agent choices while still maintaining the desired cyclical dynamics.

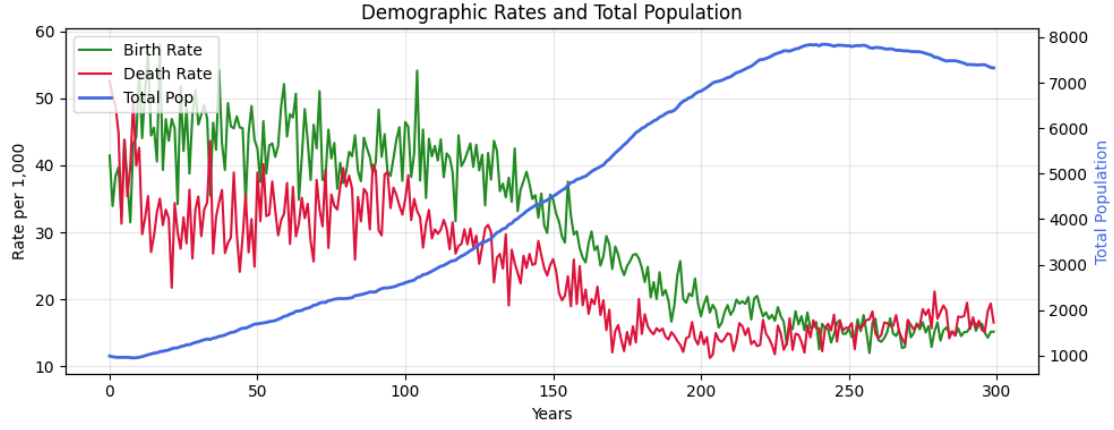


Figure 9: Demographic Rates and Total Population of the Fashion Trend Model with Population Dynamics.

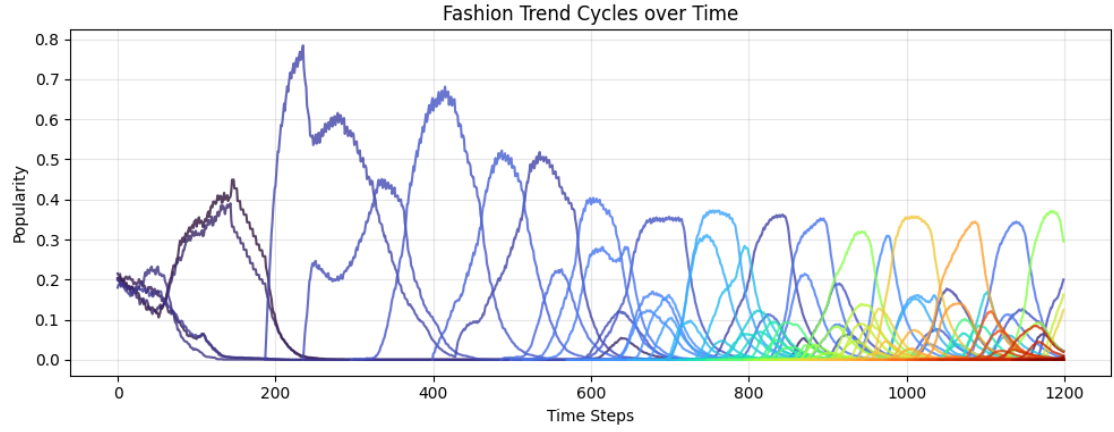


Figure 10: Fashion Trend Cycles over Time of the Fashion Trend Model with Population Dynamics.

Overall, patterns highly analogous to those of the standalone Fashion Trend Model, which are described in Section 3.1. These patterns can be observed across Figures 11, 12, and 13. This confirms that the fusion of the two models successfully preserves the foundational mechanisms of trend adoption, passive drift, and revival, while successfully introducing population dynamics.

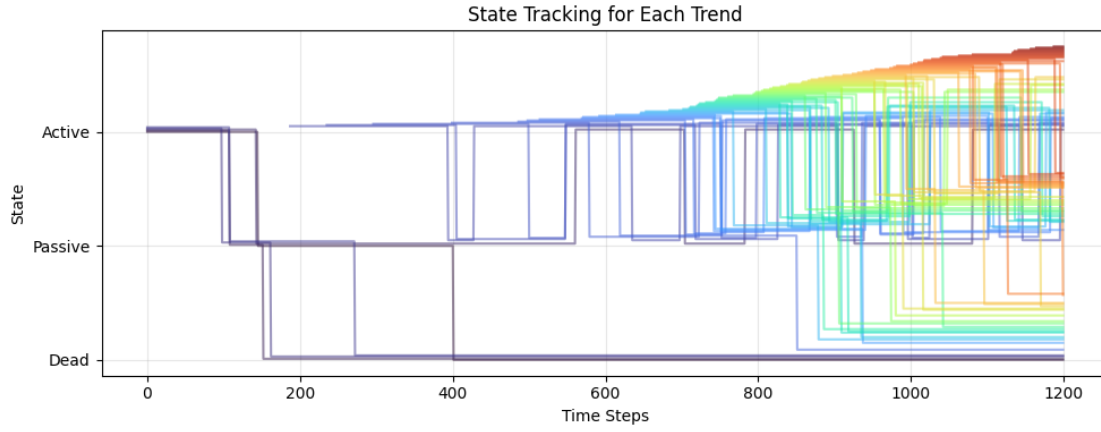


Figure 11: State Tracking for each Trend over time of the Fashion Trend Model with Population Dynamics (for better visibilty, each state curve is slightly shifted).

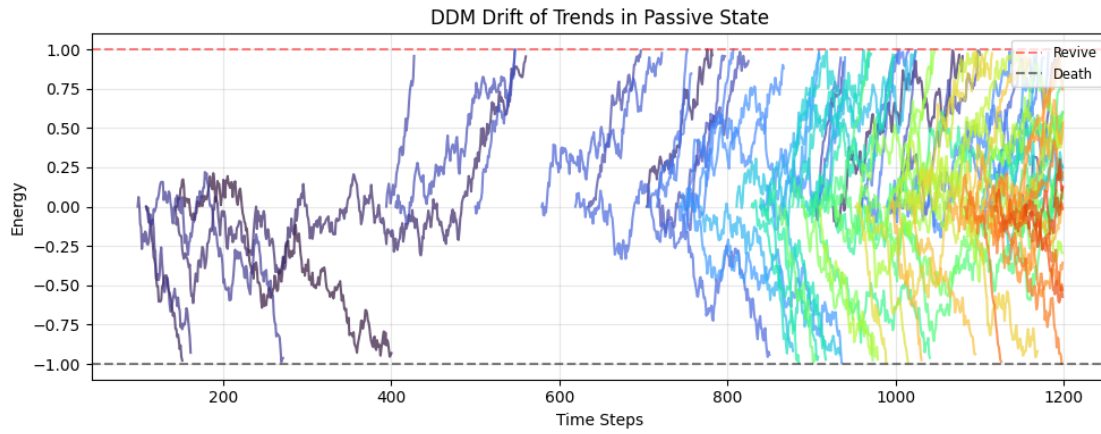


Figure 12: DDM Drift of Trends in Passtive State of the Fashion Trend Model with Population Dynamics.

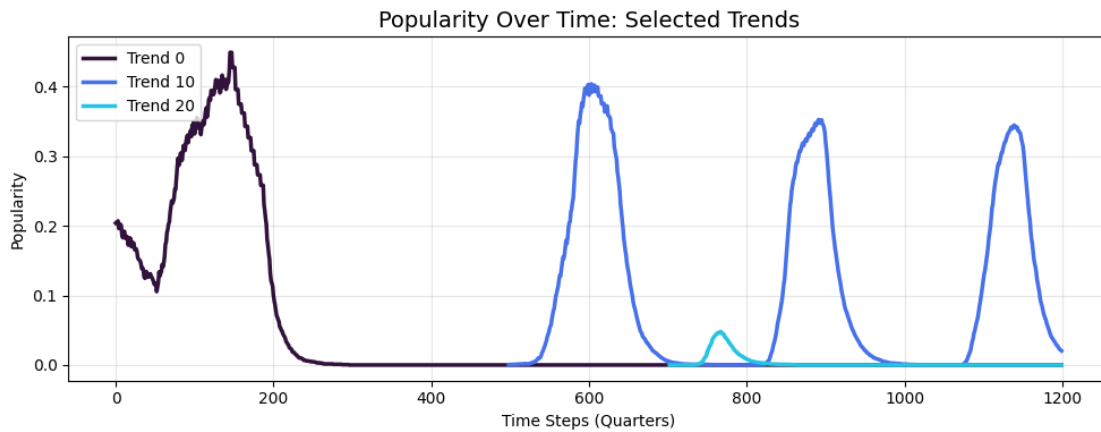


Figure 13: Popularity of selected Trends over time of the Fashion Trend Model with Population Dynamics.

4 Discussion

The results derived from the models described in the Methods section indicate that the complex, cyclical dynamics of fashion trends can indeed be successfully captured by a computational model. The standalone Fashion Trend Model generates realistic cyclical behavior, where trends become popular before decaying over time. Furthermore, the integration of the drift-diffusion model allows revival of trends while others permanently die out. Crucially, upon fusing this baseline model with the population dynamics to simulate a more realistic to the current population dynamics, the framework maintains its foundational cyclical dynamics while successfully incorporating the generational turnover. Despite these successes, the current framework is subject to an unrealistic setting and overcomplication.

A primary limitation of the current model is the assumption that an agent can only adopt a single trend at any given time. While this might hold true in historical eras where single fashion movements were present, it fails to capture modern consumer behavior, where individuals simultaneously participate in a multitude of overlapping micro-trends. Additionally, the simulation’s scale is currently restricted to a initial population of $N = 1000$ agents. While this serves as a functional proof of concept, future iterations should scale these parameters to yield more robust results and insights, which simultaneously require more computational power. Furthermore, the model’s current design prevents the emergence of classics, such as denim jeans, that start as trends but eventually become permanent wardrobe staples.

Another point of criticism is that the model itself might be too complex. It can be argued that the trends cyclic behaviour is simply a product of them being incorporated into the model. This hypothesis is supported by the forced exponential decay of appeal and the DDM mechanics used to reactivate dormant trends. Therefore, the cycles arise by design rather than naturally from the agents’ choices. Initial exploratory simulations suggest that a framework driven solely by a drift diffusion model, without the forced appeal decay, might be sufficient to replicate fashion dynamics. This leaner approach could demonstrate that trend cyclicity can emerge organically from basic stochastic drift. Future research should also explore whether a much simpler model solely from the agents behaviours could achieve the desired dynamics.

Nevertheless, these limitations define clear avenues for future work. The current unified framework provides a robust foundational sandbox that can be easily expanded upon. Ultimately, it successfully demonstrates that a targeted selection of relatively simple computational equations can effectively simulate the cyclical behavior of fashion trends within a shifting demographic landscape.

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